

VANSHEE INFOTECH

Restaurant & Hotel POS System

Software Requirements Specification (SRS)

Enterprise Edition - Version 2.0

Version History

Version	Date	Prepared By	Description
1.0	10-Jul-2026	Vanshee Infotech	Initial Release
1.1		Vanshee Infotech	UI Improvements
2.0		Vanshee Infotech	Enterprise Documentation

Document Information

Project	Restaurant & Hotel POS System
Document	SRS
Version	2.0
Company	Vanshee Infotech
Confidentiality	Internal Use Only

Table of Contents

- Introduction
- Project Overview
- Objectives
- Scope
- User Roles
- Admin Dashboard
- Customer Management
- Table Management
- QR Ordering
- Menu Management

1. Introduction

Purpose

The Restaurant & Hotel POS System automates restaurant and hotel operations including order management, billing, inventory, CRM, reporting, vendor management and hotel reservations.

Objectives

- Reduce manual work
- Improve billing accuracy
- Manage inventory efficiently
- Support QR Ordering
- Provide real-time dashboards

Scope

Supports restaurants, cafes, hotels, resorts, cloud kitchens and multi-branch hospitality businesses.

2. Project Overview

Overview

The system integrates Restaurant POS, Hotel Management, QR Ordering, Inventory, Vendor Management and Analytics into one centralized platform.

Business Workflow

Customer → Reservation / Walk-in → Order → Kitchen Display System → Billing → Payment → Reports

Main Modules

- Restaurant POS
- Hotel Management
- Inventory
- Vendor Management
- CRM
- Reports & Analytics

3. User Roles

Admin

Complete system control.

Manager

Manage staff and reports.

Cashier

Billing and payments.

Waiter

Create and manage orders.

Chef

Update kitchen order status.

Receptionist

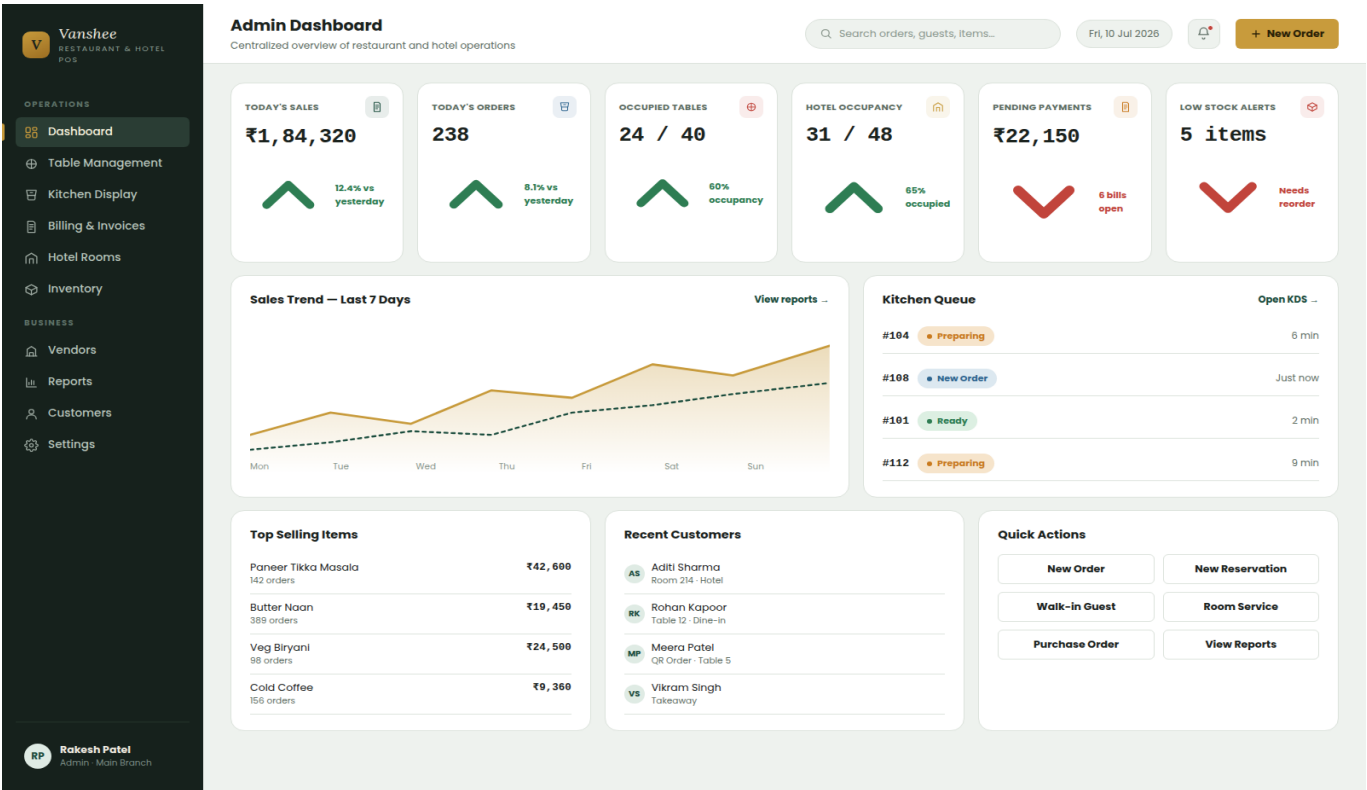
Reservation and check-in/check-out.

4. Admin Dashboard

Purpose: Centralized overview of restaurant and hotel operations.

Features

- Today's Sales
- Today's Orders
- Occupied Tables
- Kitchen Queue
- Hotel Occupancy
- Pending Payments
- Low Stock Alerts
- Quick Actions



Description: Displays overall business performance and provides quick navigation to key modules.

PART 2 – Customer Management to Payment

5. Customer Management

Purpose

Manage customer profiles, loyalty, contact details, visit history and billing information.

Features

- Add / Edit / Delete Customer
- Search by Mobile Number
- Loyalty Points
- Customer Purchase History
- Birthday & Anniversary Offers
- WhatsApp / SMS Notifications

Business Rules

- Mobile number must be unique.
- Loyalty points are credited only after successful payment.
- Duplicate customers should not be created.

Workflow

Customer Registration → Order → Billing → Payment → Loyalty Update

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Vanshee POS

HOTEL & RESTAURANT

OPERATIONS

Dashboard

Tables

Kitchen Display

Billing

Menu Management

HOSPITALITY

Rooms

Inventory

Customers

Vendors

INSIGHTS

Reports

Settings

RS

Rhea Shah

Restaurant Manager

Customers & CRM

2,948 guests · 214 active loyalty members

Search by name or phone

+ Add Customer

TOTAL GUESTS

2,948

LOYALTY MEMBERS

214

REPEAT RATE

34%

AVG. RATING

4.6

All Guests

VIP

Regular

New

Hotel Guests

GUEST	SEGMENT	VISITS	LOYALTY PTS	LAST VISIT	RATING	TOTAL SPEND
AK Aditya Kapoor +91 98200 1234	VIP	28	4,128	Today	★★★★★	₹1,84,200
MS Meera Shah +91 97250 88123	Regular	12	1,860	2 days ago	★★★★☆	₹62,400
RP Rohit Patel +91 90999 44210	New	1	80	Yesterday	★★★★☆	₹3,240
NJ Neha Joshi +91 99040 55671	VIP	41	6,530	Today	★★★★★	₹2,41,750
SV Sameer Vora +91 96380 12984	Regular	7	940	5 days ago	★★★★☆	₹28,900
PD Priya Desai +91 98980 76543	New	2	150	Today	★★★★☆	₹7,680

6. Table Management

Purpose

Manage restaurant tables and their live status.

Features

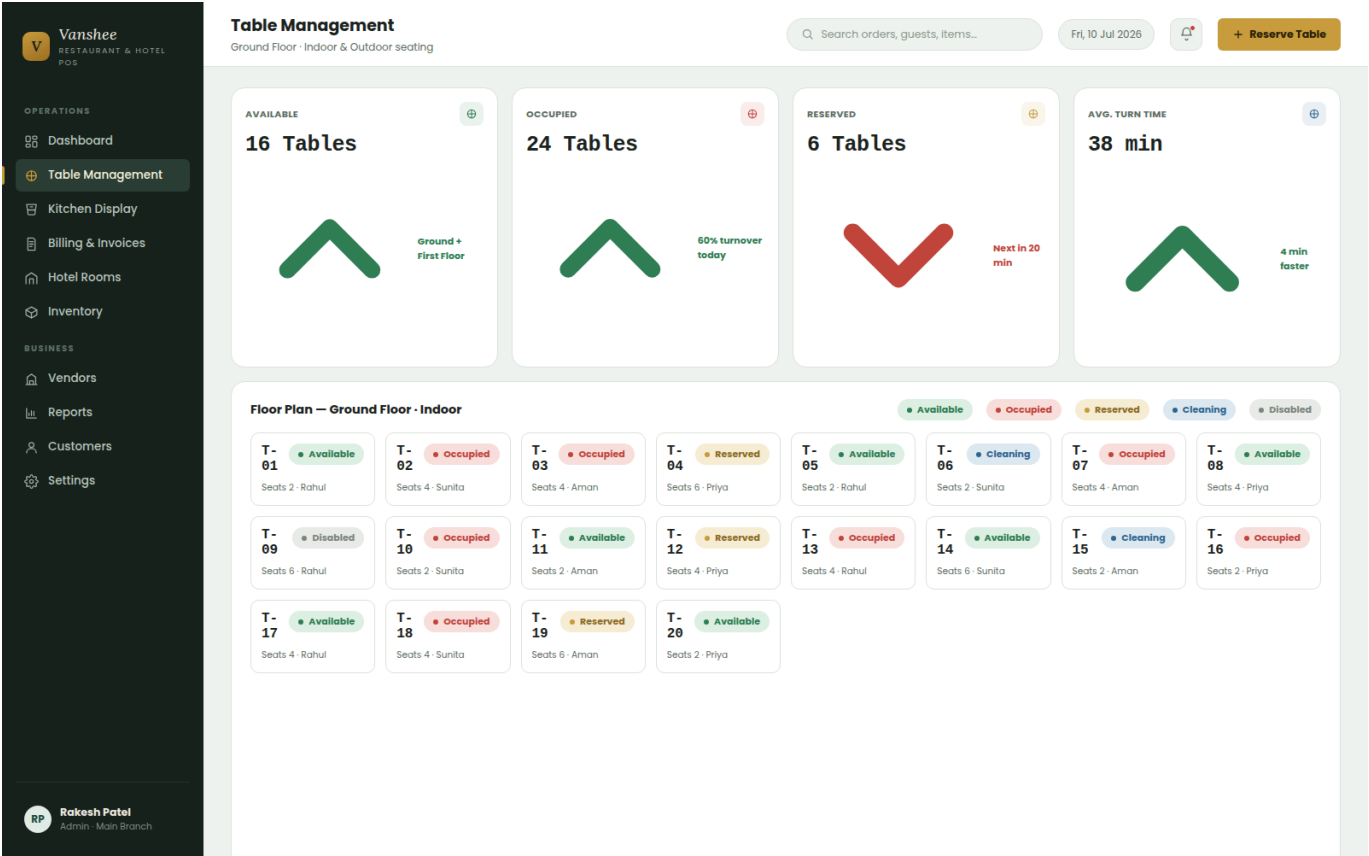
- Reserve Table
- Merge / Split Tables
- Transfer Table
- QR Code per Table
- Assign Waiter

Statuses

- Available
- Occupied
- Reserved
- Cleaning
- Disabled

Workflow

Create Table → Allocate → Order → Billing → Free Table



7. QR Ordering

Purpose

Allow customers to scan a QR code and place orders without staff assistance.

Features

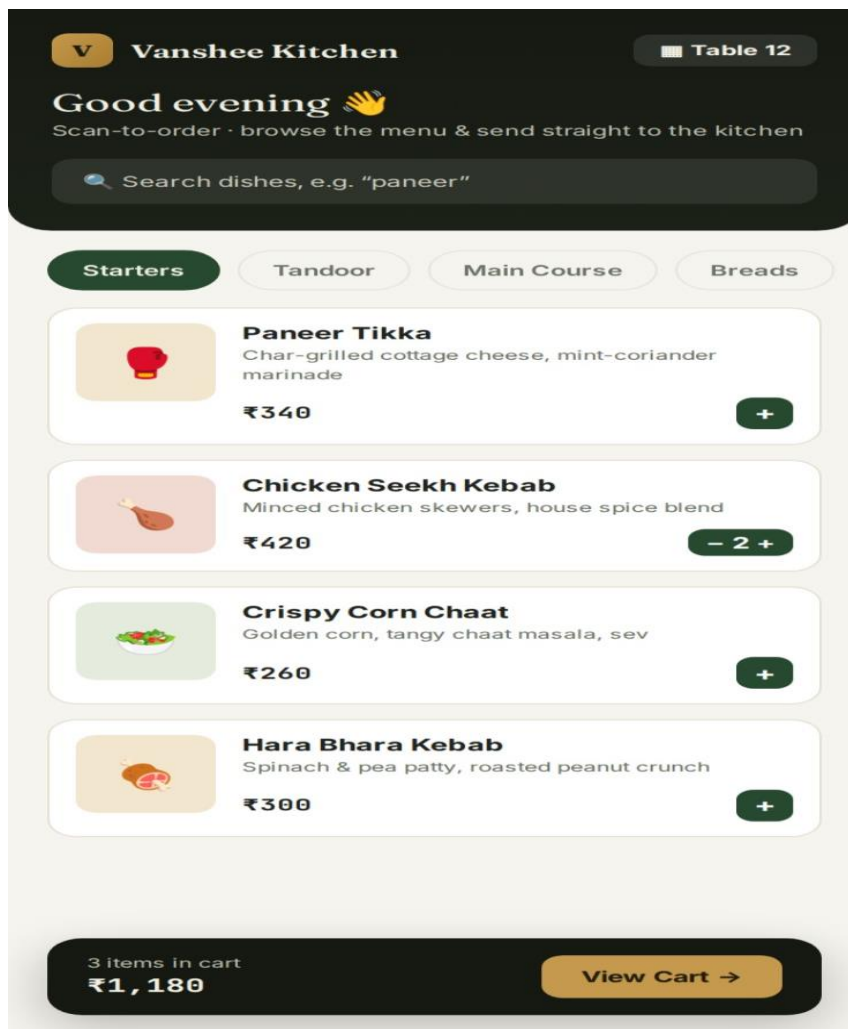
- Digital Menu
- Add to Cart
- Special Instructions
- Live Order Status
- Table Identification

Workflow

Scan QR → View Menu → Add Items → Confirm Order → Kitchen → Billing

Business Rules

- Inactive tables cannot place orders.
- Unavailable items remain hidden.



8. Menu Management

Features

- Categories
- Variants
- Add-ons
- Images
- Preparation Time
- Kitchen Department
- GST Configuration

Business Rules

- Price updates apply to new orders only.
- Hidden items are not visible in QR Ordering.

Vanshee POS
HOTEL & RESTAURANT

Menu Management
128 items across 9 categories · Restaurant menu

Search menu items | Import CSV | + Add Item

OPERATIONS

- Dashboard
- Tables
- Kitchen Display
- Billing
- Menu Management**

HOSPITALITY

- Rooms
- Inventory
- Customers
- Vendors

INSIGHTS

- Reports
- Settings

RS Rhea Shah
Restaurant Manager

Starters 18

- Soups & Salads 11
- Tandoor 14
- Main Course 26
- Breads 9
- Rice & Biryani 12
- Desserts 10
- Beverages 16
- Room Service Only 12

Paneer Tikka ₹340

Char-grilled cottage cheese, mint-coriander marinade
4 modifiers

Chicken Seekh Kebab ₹420

Minced chicken skewers, house spice blend
2 modifiers

Crispy Corn Chaat ₹260

Golden corn, tangy chaat masala, sev
No modifiers

Hara Bhara Kebab ₹300

Spinach & pea patty, roasted peanut crunch
1 modifier

Fish Amritsari ₹460

Batter-fried basa, ajwain & carom seed
3 modifiers

Mushroom Galouti ₹320

Melt-in-mouth mushroom kebab, warm spices
2 modifiers

9. Kitchen Display System (KDS)

Features

- Live Kitchen Queue
- Order Priority
- Preparation Timer
- Chef Assignment
- Status Updates

Workflow

New → Accepted → Preparing → Ready → Served → Completed

Vanshee POS
HOTEL & RESTAURANT

Operations

- Dashboard
- Tables
- Kitchen Display**
- Billing
- Menu Management

Hospitality

- Rooms
- Inventory
- Customers
- Vendors

Insights

- Reports
- Settings

Kitchen Display — Order Ticket
KOT #1042 · Table 7 · Hot Kitchen station

KOT #1040 Table 3
2x Butter Chicken
1x Garlic Naan
04:12 **Ready**

KOT #1042 Table 7
1x Paneer Tikka
2x Chicken Seekh Kebab
1x Hara Bhara Kebab
11:47 **Cooking**

KOT #1043 Room 214
1x Veg Biryani
1x Raita
18:03 **Delayed**

KOT #1044 Table 12
3x Masala Chai
01:22 **New**

KOT #1045 Table 5
1x Fish Amritsari
2x Tandoori Roti
08:38 **New**

KOT #1041 Table 9
1x Mushroom Galouti
06:55 **Ready**

KOT #1042 Table 7 · 4 guests
Fired at 8:14 PM
Elapsed 11m 47s
Server Anil Meher

1x Paneer Tikka
No onion, extra mint chutney
Tandoor

2x Chicken Seekh Kebab
Spicy level: high
Tandoor

1x Hara Bhara Kebab
No modifiers
Hot Kitchen

Mark Ready **Bump Item**

SK Suresh Kumar
Head Chef

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RESTAURANT & HOTEL

POS

OPERATIONS

Dashboard

Table Management

Kitchen Display

Billing & Invoices

Hotel Rooms

Inventory

BUSINESS

Vendors

Reports

Customers

Settings

RP

Rakesh Patel

Admin - Main Branch

Kitchen Display System

Real-time order queue for kitchen staff

Q Search orders, guests, items...

Fri, 10 Jul 2026

🔔

👤

+ Priority Order

New Orders	Preparing	Ready for Pickup
#118 Table 07 - 1 min elapsed Chicken Biryani ×2 Raita ×1 New	#114 Table 04 - 6 min elapsed Veg Biryani ×1 Raita ×2 Preparing	#111 Table 15 - 12 min elapsed Butter Chicken ×1 Naan ×2 Ready
#119 Table 12 - 1 min elapsed Paneer Tikka ×1 Butter Naan ×3 New	#115 Table 09 - 9 min elapsed Masala Dosa ×2 Preparing	#112 Table 02 - 10 min elapsed Fried Rice ×2 Ready
#120 Table QR-05 - 0 min elapsed Cold Coffee ×2 New	#116 Table 21 - 4 min elapsed Tandoori Roti ×6 Dal Makhani ×2 Preparing	

10. Billing & Payment

Billing Features

- Split Bill
- Merge Bill
- Print Invoice
- WhatsApp Bill
- Email Invoice
- Reprint Invoice

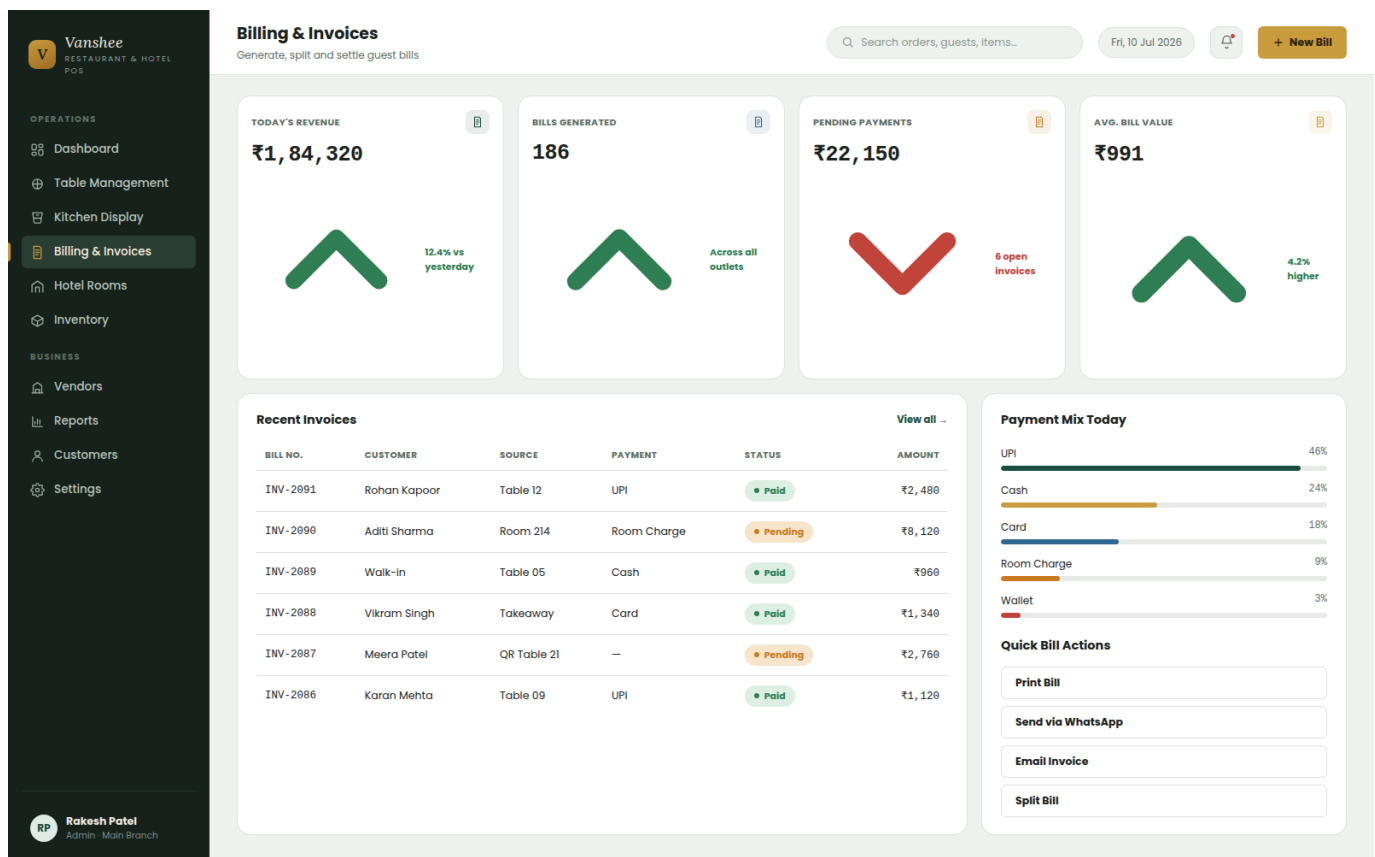
Payment Methods

- Cash
- Card
- UPI
- Wallet
- Gift Card
- Room Charge
- Multi Payment

Validation

- Digital payments require Transaction ID.
- Refunds require manager approval.

Workflow: Order → Bill → Payment → Invoice → Reports





Vanshee Hotel & Restaurant

42 Riverside Road, Ahmedabad, GJ 380015
GSTIN: 24AACC V1234F1Z5

TAX INVOICE

Invoice No. **INV-8841**

Date **10 Jul 2026, 9:42 PM**

Table **07** · Covers **4**

BILLED TO

Aditya Kapoor

+91 98200 11234

Loyalty ID: VN-2291

SERVED BY

Anil Meher

Server ID: SV-014

Restaurant · Dine-in

ITEM	QTY	RATE	AMOUNT
Paneer Tikka	1	340.00	340.00
Chicken Seekh Kebab	2	420.00	840.00
Hara Bhara Kebab	1	300.00	300.00
Butter Chicken	1	460.00	460.00
Garlic Naan	3	70.00	210.00
Masala Chai	4	90.00	360.00

Subtotal ₹2,510.00

CGST (2.5%) ₹62.75

SGST (2.5%) ₹62.75

Service Charge (5%) ₹125.50

Total ₹2,761.00

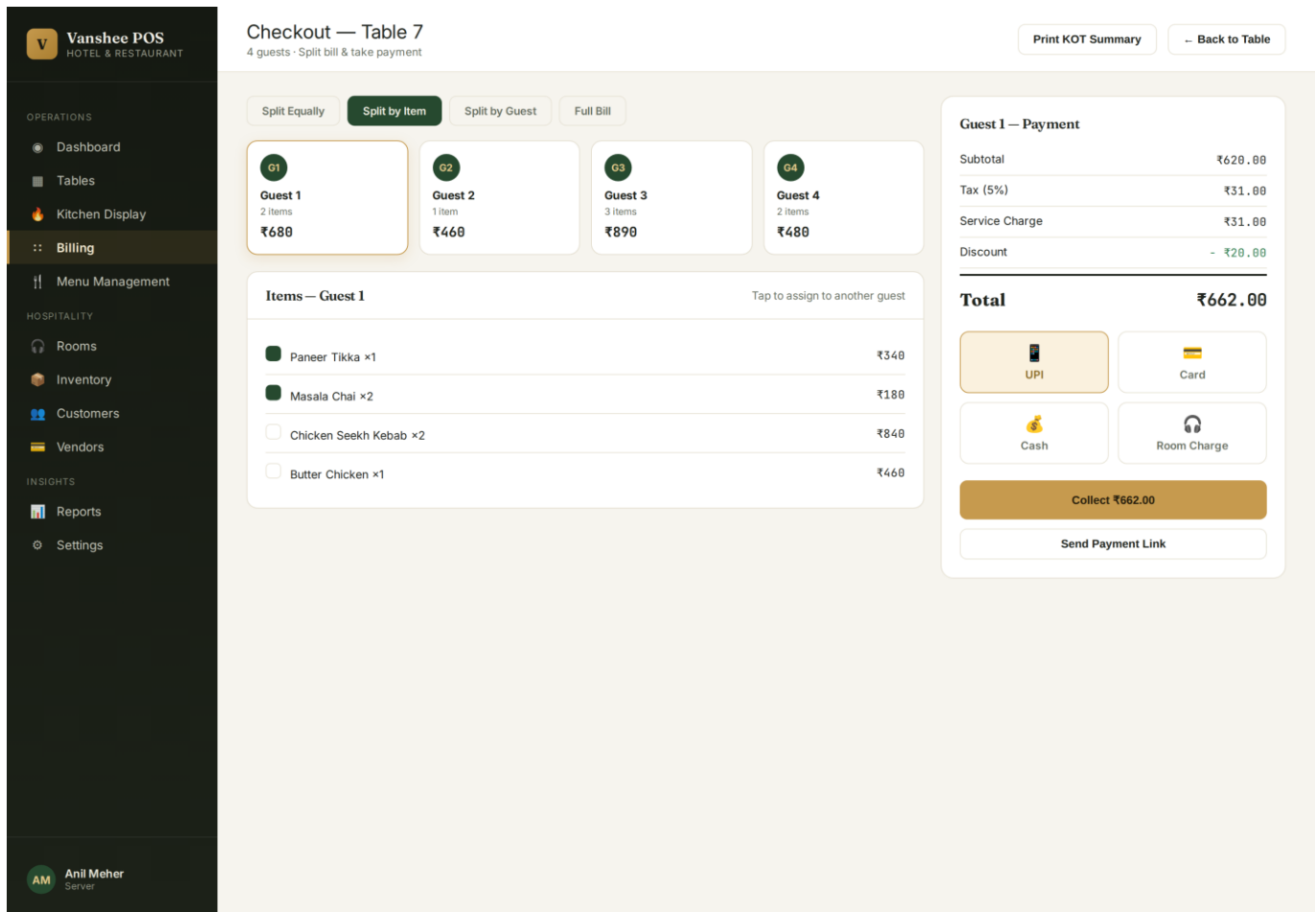


Scan to pay via UPI

vanshee@okhdfc

● Paid · UPI

Thank you for dining with us · vanshee.com · +91 79 4012 5566



PART 3 – Inventory, Hotel & Technical Documentation

11. Inventory Management

Purpose

Manage stock, recipes, suppliers and inventory movement in real time.

Features

- Stock In / Stock Out
- Stock Adjustment
- Recipe Consumption
- Low Stock Alerts
- Expiry Alerts
- Barcode & SKU Management
- Batch Tracking

- Physical Stock Count

Workflow

Purchase → GRN → Inventory Update → Consumption → Stock Report

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RESTAURANT & HOTEL

POS

OPERATIONS

Dashboard

Table Management

Kitchen Display

Billing & Invoices

Hotel Rooms

Inventory

BUSINESS

Vendors

Reports

Customers

Settings

RP

Rakesh Patel

Admin - Main branch

Inventory Management

Stock levels, reorder points and vendor supply

Q Search orders, guests, items...

Fri, 10 Jul 2026

+ Stock In

TOTAL SKUS

342

Across 6 categories

LOW STOCK ITEMS

12

Needs reorder

EXPIRING IN 7 DAYS

4 items

Perishables

STOCK VALUE

₹6.2L

Purchase price basis

Stock Overview

+ Stock In

ITEM	SKU	CATEGORY	CURRENT	MINIMUM	STATUS
Basmati Rice	KIT-0021	Grains	12 kg	20 kg	Low Stock
Paneer	KIT-0045	Dairy	8 kg	10 kg	Low Stock
Chicken (Fresh)	KIT-0032	Meat	34 kg	15 kg	Sufficient
Tomato	KIT-0102	Vegetables	22 kg	15 kg	Sufficient
Cooking Oil	KIT-0058	Essentials	40 L	25 L	Sufficient
Paper Napkins	SUP-0011	Supplies	6 packs	20 packs	Low Stock

Reorder Suggestions

Basmati Rice

8 kg short

Create PO

Paneer

2 kg short

Create PO

Paper Napkins

14 packs short

Create PO

Vendor Dues

Fresh Farms Supplies

₹18,200

Metro Dairy Co.

₹6,450

12. Vendor & Purchase Management

Features

- Vendor Registration
- Purchase Orders
- Approval Workflow
- Goods Receipt Note (GRN)
- Vendor Ledger
- Outstanding Balance

Business Rules

- GST validation for vendors
- PO approval before GRN
- Automatic stock update after GRN

Workflow: Vendor → PO → Approval → GRN → Inventory → Payment

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HOTEL & RESTAURANT

OPERATIONS

Dashboard

Tables

Kitchen Display

Billing

Menu Management

HOSPITALITY

Rooms

Inventory

Customers

Vendors

INSIGHTS

Reports

Settings

RS

Rhea Shah

Restaurant Manager

Vendor Management

32 vendors · 6 purchase orders pending approval

Search vendors

+ New Purchase Order

FF

Fresh Farms Co-op

Produce & Vegetables

Outstanding ₹12,400

Last Order 2 days ago

Lead Time 1 day

Active

MP

Metro Poultry & Meats

Meat & Poultry

Outstanding ₹38,900

Last Order Today

Lead Time 1 day

PO Pending

SD

Sundar Dairy Supplies

Dairy & Beverages

Outstanding ₹0

Last Order 4 days ago

Lead Time 2 days

Active

Purchase Orders

View All

PO #	VENDOR	CATEGORY	ITEMS	AMOUNT	ORDER DATE	PAYMENT STATUS
P0-2201	Fresh Farms Co-op	Produce	18 items	₹12,400	08 Jul 2026	Paid
P0-2202	Metro Poultry & Meats	Meat & Poultry	9 items	₹38,900	10 Jul 2026	Awaiting Approval
P0-2203	Sundar Dairy Supplies	Dairy	14 items	₹9,650	06 Jul 2026	Paid
P0-2204	Golden Spice Traders	Grocery & Spices	22 items	₹16,220	09 Jul 2026	Overdue
P0-2205	Coastal Linens & Housekeeping	Housekeeping	6 items	₹24,800	05 Jul 2026	Partially Paid

13. Hotel Management

Modules

- Room Reservation
- Check-In
- Check-Out
- Room Allocation
- Room Service
- Housekeeping
- Laundry
- Mini Bar
- Night Audit

Room Status

- Available
- Occupied
- Reserved
- Cleaning
- Maintenance
- Out of Service

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RESTAURANT & HOTEL

POS

OPERATIONS

Dashboard

Table Management

Kitchen Display

Billing & Invoices

Hotel Rooms

Inventory

BUSINESS

Vendors

Reports

Customers

Settings

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Rakesh Patel

Admin - Main Branch

Hotel Room Management

Reservations, check-in/out and housekeeping

Q Search orders, guests, items...

Fri, 10 Jul 2026

+ New Reservation

OCCUPANCY

65%

31 of 48 rooms

TODAY'S CHECK-INS

9

3 pending arrival

TODAY'S CHECK-OUTS

6

2 completed

ADR

₹4,250

Average daily rate

HOUSEKEEPING DUE

5 rooms

cleaning in progress

Room Grid — Floor 2

Available

Occupied

Reserved

Cleaning

Maintenance

2001 Deluxe Occupied	2002 Deluxe Available	2003 Suite Available	2004 Standard Reserved
2005 Deluxe Cleaning	2006 Deluxe Occupied	2007 Suite Available	2008 Standard Occupied
2009 Deluxe Available	2010 Deluxe Maintenance	2011 Suite Occupied	2012 Standard Available
2013 Deluxe Available	2014 Deluxe Reserved	2015 Suite Occupied	2016 Standard Available

Arrivals — Next 3 Hours

Aditi Sharma	2:00 PM
Room 214 - Deluxe	
Farhan Ali	3:30 PM
Room 306 - Suite	
Neha Joshi	4:15 PM
Room 108 - Standard	

Guest Ledger Snapshot

Room Charges	₹18,400
Room Service	₹3,120
Laundry	₹640

Reservation Calendar

10-19 Jul 2026 · 22 rooms · 78% occupancy

← Prev

Next →

+ New Booking

Room	10 Fri	11 Sat	12 Sun	13 Mon	14 Tue	15 Wed	16 Thu	17 Fri	18 Sat	19 Sun
201 Deluxe King	Kapoor family					Neha Joshi				
202 Deluxe King		Corporate - Infosys								210 Executive Suite
Maint.		Rohit Patel				Sameer Vora				214 Deluxe Twin
		Priya Desai	Priya Desai				Meera Shah			301 Presidential Suite
			Sharma Wedding Block							

Confirmed booking

Group / corporate block

Guest checking in today

Maintenance / blocked

15. Database Design

Core Tables

- users
- roles
- customers
- tables
- menu_items
- orders
- order_items
- kitchen_orders
- bills
- payments
- inventory
- vendors
- purchase_orders
- rooms
- reservations
- employees
- attendance
- notifications

16. Functional Requirements

- FR-001 User Login
- FR-002 Customer Registration
- FR-003 Table Reservation
- FR-004 QR Ordering
- FR-005 KOT Generation
- FR-006 Bill Generation
- FR-007 Split Bill
- FR-008 Multi Payment
- FR-009 Vendor Management
- FR-010 Inventory Update
- FR-011 Room Reservation
- FR-012 Reports

17. Security & Future Enhancements

Security

- Role-Based Access Control
- Encrypted Passwords
- Audit Logs
- Session Timeout
- Daily Database Backup

Future Enhancements

- AI Sales Forecasting
- Voice Ordering
- WhatsApp Ordering
- Self-Service Kiosk
- Swiggy Integration
- Zomato Integration
- Customer Mobile App
- Multi-Branch Analytics

PART 4 – Enterprise Modules & Final Documentation

18. Login & Authentication

Purpose

Secure authentication for all users with role-based access.

Features

- Role-Based Login
- Forgot Password
- OTP Verification
- Remember Me
- Session Timeout
- Two-Factor Authentication (Optional)

Workflow

Login → Verify Credentials → Dashboard



Vanshee POS
HOTEL & RESTAURANT SUITE

One system for the front desk, the floor, and the kitchen.

Sign in to manage tables, rooms, orders and billing — all from a single dashboard built for hospitality teams.

22

ROOMS MANAGED

3,412

ORDERS / MONTH

99.9%

UPTIME

Welcome back

Sign in to your Vanshee POS staff account



Staff



Manager



Admin

Username or Staff ID

rhea.shah

Password

••••••••

☒ Remember this device

[Forgot password?](#)

Sign In

or continue with



PIN Login



QR Badge Scan

Having trouble signing in? Contact your outlet manager or IT support at support@vanshee.com



Vanshee POS
HOTEL & RESTAURANT SUITE

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UPTIME

Welcome back

Sign in to your Vanshee POS staff account



Staff



Manager



Admin

Username or Staff ID

rhea.shah

Password

••••••••

☒ Remember this device

[Forgot password?](#)

Sign In

or continue with



PIN Login



QR Badge Scan

Having trouble signing in? Contact your outlet manager or IT support at support@vanshee.com



Reset your password

Enter the email or phone number linked to your staff account. We'll send a 6-digit verification code to reset your password.

Registered Email or Phone

rhea.shah@vanshee.com

Cancel

Send Reset Code

19. Employee Management

Features

- Employee Registration
- Attendance
- Shift Management
- Leave Management
- Salary Records
- Performance Tracking

Benefits: Better staff management and productivity.

20. Roles & Permissions

- Admin
- Manager
- Cashier
- Waiter
- Chef
- Receptionist
- Inventory Manager
- Housekeeping

Each role has dedicated access rights and module permissions.

21. API Integrations

- Razorpay Payment Gateway
- PhonePe UPI
- Google Pay
- WhatsApp API
- SMS Gateway
- Swiggy Integration
- Zomato Integration
- Google Maps

22. Hardware Support

- Thermal Printer
- Barcode Scanner
- QR Scanner
- Cash Drawer
- Customer Display
- Kitchen Display Screen
- Biometric Device

23. Non-Functional Requirements

- High Availability
- Fast Response Time
- Scalable Architecture
- Secure Authentication
- Daily Backup
- Role-Based Security
- Cross Platform Support

24. Testing Scenarios

- Login Testing
- Billing Testing
- Inventory Testing
- QR Ordering Testing
- Payment Testing
- Reservation Testing
- Performance Testing
- Security Testing

